

Welcome! We will begin shortly



## Live Virtual Class

### The Must-Know Mathematics & Statistics Behind AI

#### Learning Outcomes

- Understand the key mathematical and statistical concepts in the World of AI
- Understand how some of different mathematical and statistical concepts are used in different scenarios
- Get questions & concerns on mathematical and statistical concepts resolved

Guidelines



Listen only mode



Ask questions at the interest of the larger audience



Questions in the Q&A Box

Thank you

Kindly utilize the chat box for **subject-relevant questions only** to maximize your learnings from the session.

**“Measure of the likelihood or chance of an event occurring”**

$$\text{Probability of an event occurring} = \frac{\text{\# outcomes associated with the event}}{\text{Total \# of outcomes}}$$



← 100% Chance (Certainty)

← 50% Chance (Equally Likely)

← 0% Chance (Impossibility)

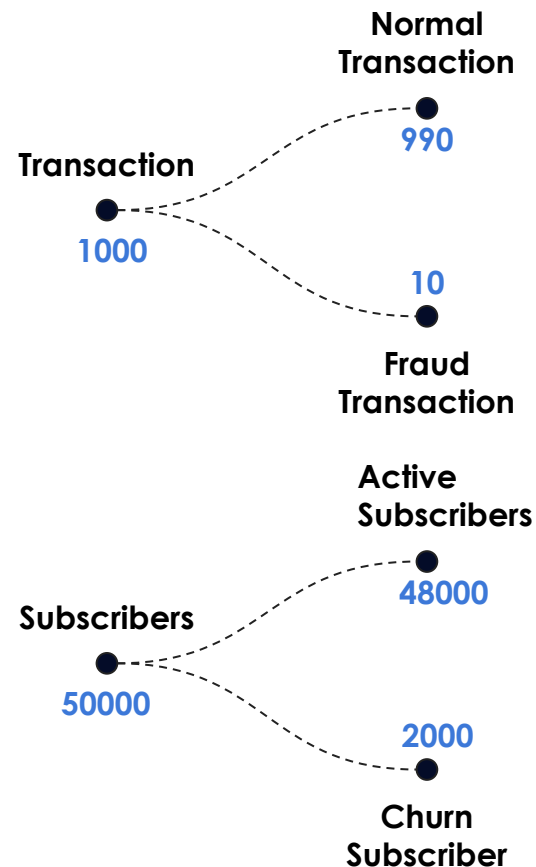
# Probability

**Example 1:** 1000 transactions, 10 are fraudulent and 990 are normal. The probability of randomly selecting a fraudulent transaction is 0.01 or 1%.

- Fraudulent cases are rare but important to detect.

**Example 2:** In a telecom company with 50,000 subscribers, 2,000 churned last month. The probability of a randomly chosen subscriber churning is 4%.

- Understanding this helps in devising retention strategies for business sustainability.



# Conditional Probability

“The probability of an event happening, given that another event has already happened”

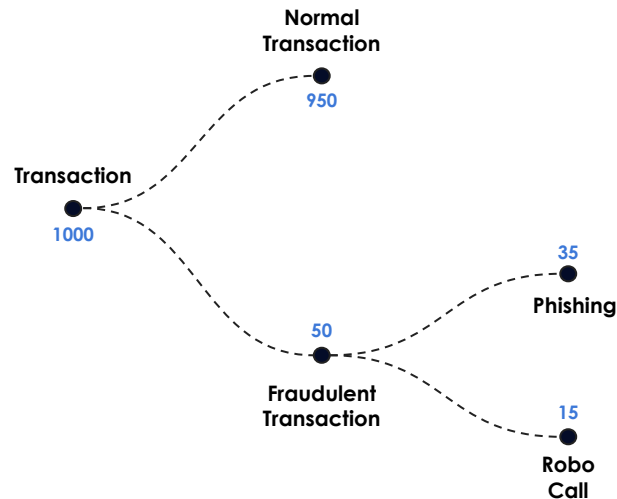
$$\text{Probability of event A happening given B} = \frac{\text{Probability of both events A and B happening}}{\text{Probability of event B happening}}$$

**Example:** Out of 1000 transactions made in an hour, 50 transactions were fraudulent. Out of those 50, 35 transactions are related to phishing scams, and 15 are related to Robo call scams.

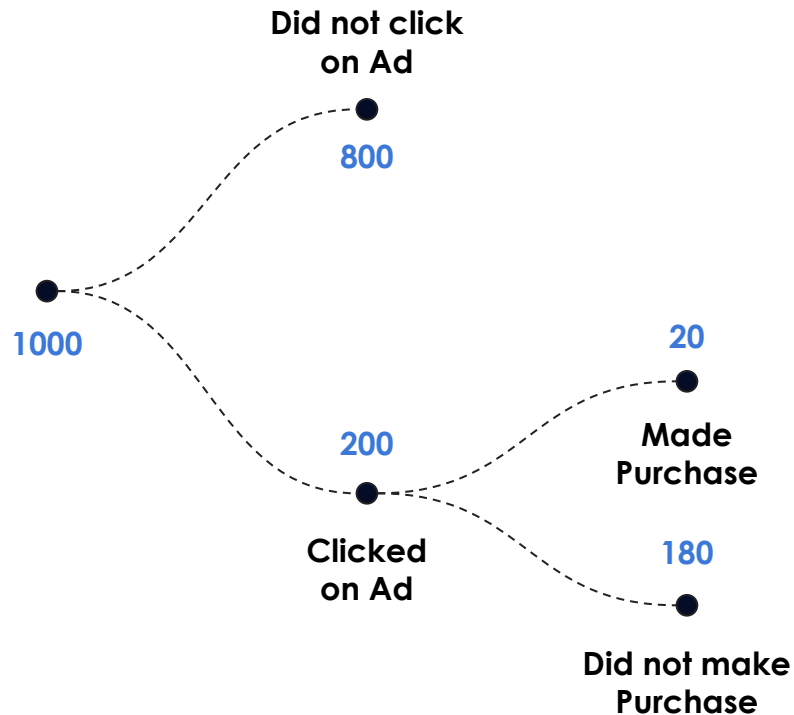
The probability of a transaction being a phishing scam given it is a fraudulent transaction can be computed as follows:

$$\begin{aligned}\text{Probability} &= \# \text{ phishing-related frauds} / \# \text{ fraudulent transactions} \\ &= 35/50 \\ &= 0.7 \text{ (or 70\%)}\end{aligned}$$

So, if we know that a transaction is fraudulent, there is a 70% chance that it is a phishing scam.



## Probability - Quiz



A company is running a marketing campaign to promote its new product. From the adjacent flowchart, compute the probability that customer who clicks on the ad will make a purchase?

A

0.2

B

0.1

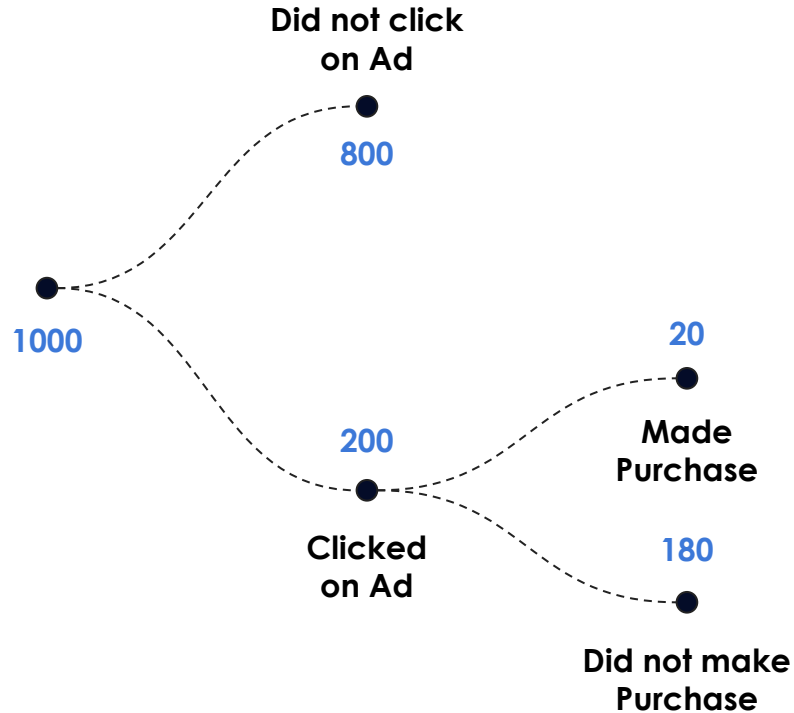
C

0.32

D

0.12

## Probability - Quiz



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0.32

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0.12

## Probability - Quiz

0.2

0.1

0.32

0.12

The question aims to emphasize that the purchase was made after clicking on the ad. Once the user clicks on the ad, there are only two possible outcomes - either they make the purchase (buy the product) or they do not make the purchase.

Out of the 200 ad clicks, 20 of them resulted in a purchase. So, we have

$$P(\text{click on ad}) = 200/1000 = 0.2$$

$$P(\text{click on ad and make a purchase}) = 20/1000 = 0.02$$

$$\begin{aligned} P(\text{make a purchase given that they clicked on the ad}) &= P(\text{click on ad and make a purchase}) / P(\text{click on ad}) \\ &= 0.02 / 0.2 \\ &= 0.1 \end{aligned}$$

Therefore, the probability of a customer clicking on the ad and then making a purchase is 0.1.

# Types of Statistics

Draw samples from the population to understand its characteristics

POPULATION



SAMPLE



Draw inferences about the population from the sample

## Inferential Statistics

Confidence Intervals - The size of the transistor on a processor chip lies in the 95% confidence interval (4.95, 5.05) nm

Hypothesis Testing - Does the conversion rate of a marketing campaign vary with the font style of the infographic?

## Descriptive Statistics

Central Tendency - Mean, Median, Mode

Dispersion - Variance, Range, Standard Deviation



# Descriptive Statistics

## Mean

A measure of the centre of the data, and is computed as the sum of all data points divided by the total number of data points

## Standard Deviation

A measure of how dispersed the data is in relation to the mean

- Low standard deviation - data is clustered around the mean
- High standard deviation - data is more spread out

### Example:

The annual salaries of 6 employees in an organization (in thousands dollars) are as follows:

40, 42, 39, 45, 48, 50

$$\begin{aligned}\text{Mean} &= \text{Sum of Salaries of Employees} / \text{Total Number of Employees} \\ &= 264 / 6 \\ &= 44\end{aligned}$$

The Mean(average) salary of employees in organizations is \$44K

### Example:

In the above scenario, the standard deviation of employee salaries comes out to be ~\$4.5K.

This is a comparatively low value, indicating that the data is clustered around the mean

# Descriptive Statistics

- In the previous scenario, a new employee joined the organization at an annual salary of \$150K
- New mean salary of employees - ~\$60K
- \$60K **not reflective of the centre of the data** - an impact of the one extremely high value
- Need a 'better' measure of the centre of the data

# Descriptive Statistics

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## Median

The middle value of the data when arranged in an order

- Odd number of data points - the actual middle number
- Even number of data points - the average of the two middle numbers

### Example:

The annual salaries of the seven employees (in thousands dollars), arranged in an order, are as follows:

39, 40, 42, 45, 48, 50, 150

Since we have an odd number (7) of data points, the median is the middle value. So, we say that the median annual salary of employees in the organization is 45

# Descriptive Statistics

- The employees whose salary we discussed previously are from the following department:  
Sales, Sales, Marketing, Sales, Marketing, HR, Finance
- What is the 'centre' of the data now?
- Can't use mean and median - **no numbers!**
- Need a 'better' measure of the centre of non-numerical data

# Descriptive Statistics

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Sales, Sales, Marketing, Sales, Marketing, HR, Finance
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## Mode

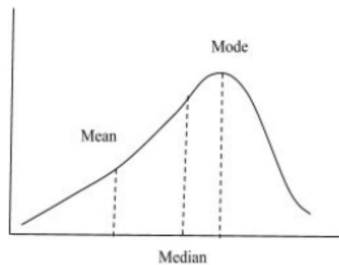
The value that occurs the most often in the data

- Data can have multiple modes
- Best used when you want to indicate the most common response or item in a data set

### Example:

In the current scenario, the most frequently occurring value is Sales  
So, the mode of the data is Sales  
This information helps business leaders understand which department forms the largest part of the workforce and they can allocate resources accordingly to ensure it functions efficiently and meets business objectives..

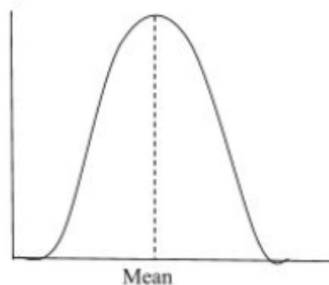
# Mean vs Median vs Mode - When to use



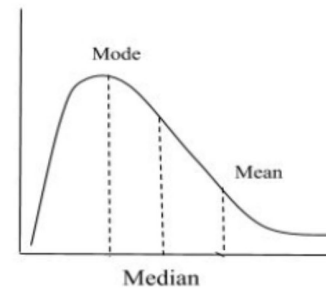
Skewed Data  
 $\text{mean} < \text{median} < \text{mode}$



**Negative Direction**



Symmetrical Data  
 $\text{mean} = \text{median} = \text{mode}$



Skewed Data  
 $\text{mode} < \text{median} < \text{mean}$

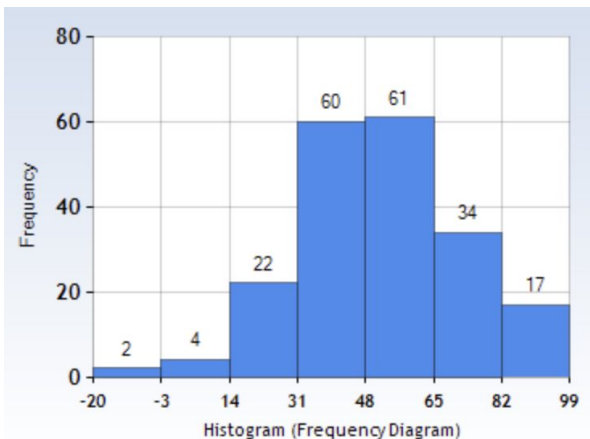


**Positive Direction**

# Statistics - Quiz

Assume that the below distribution was created using 200 data points. The mean of this distribution is 50. The median of this dataset will also be roughly the same.

If you add another 100 data points with values between 60 and 75, what will the relation between the mean & the median look like?



A

Mean > Median

B

Mean < Median

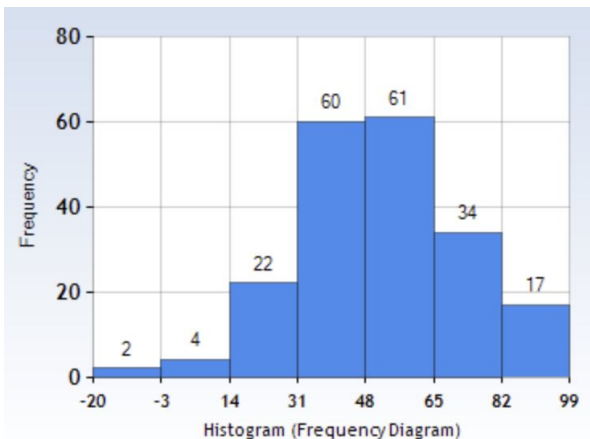
C

Mean = Median

# Statistics - Quiz

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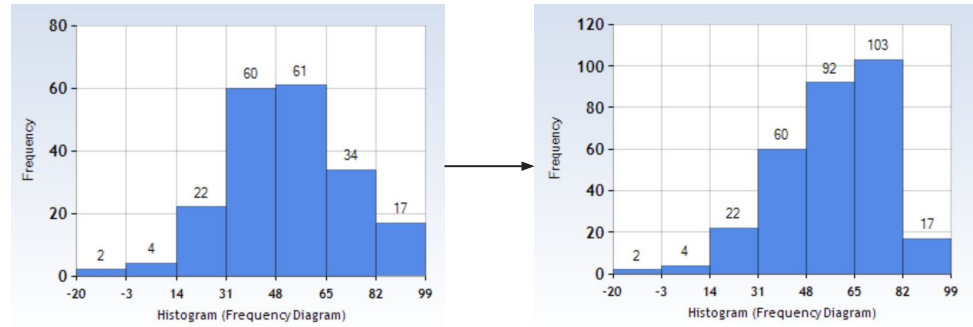
# Statistics - Quiz

Mean > Median

Mean < Median

Mean = Median

While the exact numbers of mean & median will depend on the actual data that was inserted, it can be inferred that with the addition of higher-valued data into the dataset, the data will lose its symmetry and be negatively skewed. This will result in the median of the data being higher than the mean.



## Statistics - Quiz

**An organisation pays \$30,000 a year to 99 of its employees, while the salary of the CEO (not included in the 99 considered before) is \$600,000 a year.**

**Assume that the organisation is hiring for a job position, which metric do you think will the organisation advertise, in order to make itself appear as a high-paying organisation?**

**A**

Mean

**B**

Median

## Statistics - Quiz

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Mean

B

Median

# Statistics - Quiz

Mean

Median

The median salary of the organisation will be roughly \$30,000. However, if you consider the mean salary of the organisation, that number will be roughly ~\$35,000.

While the figures may appear impressive at first glance, they do not provide a complete picture as one of the employees (who happens to be the boss) is earning considerably more than the other workers. As a result, such outcomes can be classified as **deceptive statistics**. This is an example of why it is usually better to check if the 'average' metric being used is median or mean, in order to be able to form a more informed opinion

“Rectangular array of numbers, symbols, or expressions arranged in rows and columns”

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 5 | 5 | 5 | 2 | 1 | 4 | 1 |
| 4 | 4 | 4 | 3 | 1 | 3 | 3 |
| 3 | 3 | 3 | 3 | 2 | 4 | 4 |
| 2 | 2 | 2 | 1 | 2 | 2 | 5 |
| 5 | 5 | 5 | 3 | 2 | 4 | 3 |

**Dimension (m $\times$ n) = 5 $\times$ 7**

**Rows (m) = 5**

**Columns (n) = 7**

# Tabular Data Representation using Matrices

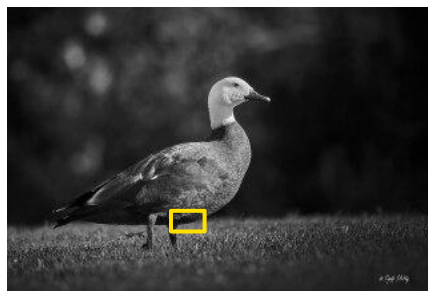
“Each row is a record (customer, object, etc.),  
each column is an attribute (age, dimensions, etc.)”

Dimension of data matrix = 5x7

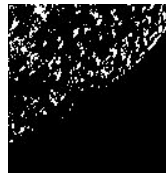
|       | Cylinders | Weight | Displacement | Horsepower | Model Year | Country | MPG |
|-------|-----------|--------|--------------|------------|------------|---------|-----|
| Car 1 | 8         | 3504   | 307          | 130        | 1970       | USA     | 18  |
| Car 2 | 4         | 2372   | 113          | 95         | 1970       | Japan   | 24  |
| Car 3 | 4         | 2130   | 97           | 88         | 1970       | USA     | 27  |
| Car 4 | 3         | 2320   | 70           | 97         | 1970       | Japan   | 19  |
| Car 5 | 3         | 2130   | 70           | 90         | 1970       | Germany | 18  |

# Image Representation using Matrices

“Convert an image into an array of numbers and each pixel has an intensity number”

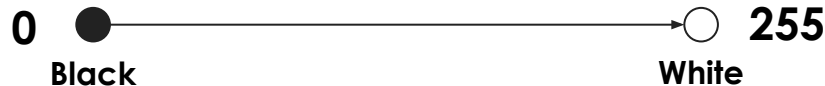


128 x 128 matrix



|     |     |     |
|-----|-----|-----|
| 010 | 146 | 025 |
| 057 | 071 | 082 |
| 012 | 098 | 099 |

9 x 9 matrix



## Image Representation - Quiz

**Why do you think the pixel intensity values range from 0 - 255?**

A

This range of values provides enough precision to represent a wide range of colors and shades, while still being computationally efficient

B

This range of values is the only range that can be represented by an 8-bit binary number

C

This range of values is the only range that can be processed by modern image processing algorithms

D

This range of values is easier to display on modern computer monitors



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## Image Representation - Quiz

This range of values provides enough precision to represent a wide range of colors and shades, while still being computationally efficient

This range of values is the only range that can be represented by an 8-bit binary number

This range of values is the only range that can be processed by modern image processing algorithms

This range of values is easier to display on modern computer monitors

This is the main reason why images are represented in memory using values from 0 to 255. This range provides enough precision to represent a wide range of colors and shades, while still being computationally efficient and easy to store in memory

While it is true that an 8-bit binary number can represent values from 0 to 255, this is not the only range of values that can be represented by an 8-bit binary number. Other ranges of values can be represented by using different formulas to map the binary values to actual values.

Modern image processing algorithms can handle images with a wide range of values, including values outside the range of 0 to 255. However, this range is often used as a standard for representing grayscale values in digital image processing

While modern computer monitors can display images with values from 0 to 255, they can also display images with higher bit depths and a wider range of values. The range of 0 to 255 is not the only range of values that can be displayed on modern computer monitors



# Thank you!

**We'd love to hear your feedback!**  
**Please share your feedback for the session**

**Wish you all the very best!**

**Please feel free to raise a [Support Request](#) through Olympus in case of any queries**